

Artificial Intelligence — 10-Page Master Quick Revision

High-Yield Formula Sheet, Search Matrices & Top BIT Mesra PYQ Solutions

🧠 10-Page Master Quick Revision — Artificial Intelligence (CS24307)

Page 1-2: PEAS Descriptions, Rationality & Environment Taxonomies

Page 5-6: Logic Entailment, CNF Skolemization & FOL Resolution

Page 9-10: Decision Tree ID3, Neural Net Backprop & Bias-Variance

Page 3-4: Search Complexity (BFS/DFS/A*), Minimax & Alpha-Beta Pruning

Page 7-8: STRIPS Planning, Bayes' Rule & Bayesian Belief Networks

⚡ High-Yield AI Formula Sheet

Algorithm / Concept	Key Formula / Principle	Complexity / Critical Property
A* Search	$f(n) = g(n) + h(n)$	Optimal for admissible $h(n) \leq h^*(n)$.
Alpha-Beta Pruning	Prune when $\alpha \geq \beta$	Reduces branching factor from b to $\sqrt{b}$ in optimal move ordering.
Bayesian Network	$P(X_1, \dots, X_n) = \prod P(X_i \mid \text{Parents}(X_i))$	D-separation determines conditional independence.
Entropy	$H(S) = - \sum p_i \log_2(p_i)$	Maximal when probabilities are uniformly distributed.